"Empowerment through quality technical education"

Ajeenkya DY Patil School of Engineering

Dr. D. Y. Patil Knowledge City, Charholi Bk., Via. Lohegaon, Pune-412 105.

Department of Computer Engineering

Assignments **Form No. IQAC/18a1**

Class: **S.E.** Div: **C**	Course Name: **Data Structures-PCC-201-COM**	Course Coordinator: **Prof.Ishwar Bharambe**
Assignment No.**01**	Assignment based on Units: **I**	Date of issue: **21 / 07 / 2026**

CO 1: Understand and Analyze various types of data structures and algorithms

Sr. No.	Questions	Level of Mapping and Number			Blooms Level
		CO	PO	PSO	
Q.1.	Explain different Asymptotic notation with graph of order growth.	CO1	PO1	PSO1	L2 – Understand
Q.2.	Define ADT. Explain Types of data structure compare Linear and Non-linear data structure	CO1	PO1	PSO1	L2 – Understand
Q.3.	Compare Static and Dynamic data structures. Discuss the significance of Persistent and Ephemeral data structures	CO1	PO1	PSO1	L2 – Understand

Q.4.	Apply Step count method to evaluate time complexity of the following python code:
	def oddEven (n=100):
	for num in range (1,n):
	if num%2 == 0:
	print(num,"is Even")
	else:
	print(num,"is Odd")
	Justify that the above Algorithms is having Linear and Non-linear, Quadratic time complexity. List various Algorithmic Strategies. Explain any one with suitable example. **[**
		CO1	PO1	PSO1	L3 – Apply

Q.5.	Differentiate between Linear and Non-linear data structure	CO1	PO1	PSO1	L2 – Understand
Sign of Course Coordinator		Sign of HoD			

